

to reduce the time for sending to the market as well as remain cost-effective throughout the life of the product.

All products typically go through a five-step process till manufacturing (most companies have specialised product stages, tuned to their development methodology) as shown in Fig. 1. The figure shows the lifecycle for any product, especially relevant for electronic products. It includes the manufacturing stage and other stages, till the end of the product's life.

The decisions taken during the early phase of the product's development decide the eventual success or failure of the product in the market. If we look at the statistics, only two out of 100 products succeed in the market, and most of the time, the two products are those in which the manufacturing issues have been already addressed.

It is important to note that while we talk about the product lifecycle, most consumer products never have a long lifecycle, which increases the pressure on the designers to make the product succeed. Companies like Sony, Panasonic and Toyota continuously launch successful products because of their design and engineering practices. These address all aspects of design and manufacturing in a holistic way, with manufacturing being part of the development process right from day one. On careful analysis, one will realise that these companies develop products that are easy to manufacture and use as well.

Going beyond the lifecycle, if we look at the product development process alone, the conventional development process typically follows what's shown in Fig. 2. Here you will notice that there are very specific process stages like supply chain management (SCM), which basically controls the cost of the product. Fig. 3 shows that 80 per cent of product cost is for material and unless that is controlled, product pricing will be a major issue later on in the market.

The mechanical design and the tooling should happen parallel to the product development, which in the

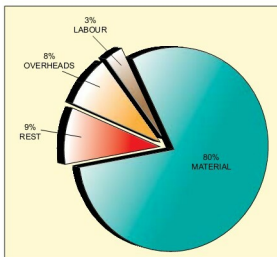


Fig. 3: Cost of elements in the manufacturing process

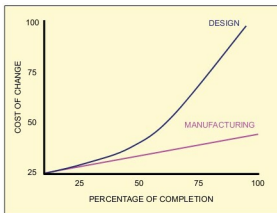


Fig. 4: Relationship between cost of changes in design and maturity of the development cycle

conventional development process happens sequentially. This effectively emphasises the need for the design and manufacturing team's coordination right from the product development stage. This becomes much more critical when the manufacturing is outsourced.

Why outsource manufacturing?

While talking about DFx, we should also appreciate the fact that worldwide, product companies are moving away from captive manufacturing and towards outsourced manufacturing. Companies that execute this outsourced manufacturing are called electronic manufacturing service (EMS) vendors.

The predominant motives for this are, firstly, globalisation of the market. Due to this, a vendor with only a few manufacturing facilities is increasingly finding it difficult to access the market in a cost-effective way due to logistics as well as access to low-cost custom parts. The second important factor is the cost of the product. EMS vendors with multiple products in their service line are able to get components at a much better price and this allows them to manage inventories much better. Above all, expensive manufacturing capital equipment is better utilised by the EMS vendors due to the load, as compared to captive manufacturing by the product companies themselves.

One of the biggest proponents and users of contract manufacturing are the consumer electronics companies. Despite their enormous volumes, which quite justify their own captive manufacturing facilities, they have opted for outsourced manufacturing due to access to global locations, which the EMS vendors offer and control cost.

When the EMS vendors set up globally-located facilities, they also create an ecosystem of vendors for custom parts in plastics, sheet metal, etc, which will be expensive if manufactured in one place and shipped to the place of manufacturing. This is especially true in the case of white goods like refrigerators, washing machines, etc, which contain large and heavy parts. Another advantage of the EMS model is that with the increased country-specific compliance standards, it becomes important to meet these standards only if the product is manufactured in the respective countries.